

# Chapter 5: Bessel Functions

## Mathematical Physics

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# Outline

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Half-integer order: spherical Bessel functions
- 8 Summary

# Where we are going

Bessel's equation of integer order  $m$  is

$$x^2 y''(x) + xy'(x) + (x^2 - m^2)y(x) = 0. \quad (5.0.1)$$

It arises when variables are separated in circular or cylindrical geometries. This chapter develops

- a generating function for the integer-order family  $J_m$ , and the power series;
- the order and derivative recurrences;
- one carefully derived real integral representation, the Schlöfli contour for integer order, and the open Sommerfeld contour for noninteger order;
- the small- and large-argument expansions (large  $x$  by stationary phase, the real oscillatory case of steepest descent from Ch. 1);
- the second solution  $Y_m$ , a short look at Hankel functions  $H^{(1)}, H^{(2)}$ , and the discrete Fourier–Bessel series on the disk;
- the half-integer orders, where the family reduces to elementary functions: the spherical Bessel and Hankel functions.

Throughout the generating-function sections,  $m, n \in \mathbb{Z}$  unless stated otherwise. For small- or large-argument limits, the order is held fixed.

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# Generating function for the family $\{J_m\}$

**Why a generating function?** Circular problems give one radial function for each integer angular order  $m$ . Package the functions into one Laurent series:

$$g(x, t) = \exp \left[ \frac{x}{2}(t - t^{-1}) \right] = \sum_{m=-\infty}^{\infty} J_m(x) t^m, \quad t \neq 0. \quad (5.1.1)$$

The notation  $[t^m]g$  means the coefficient of  $t^m$ , so  $J_m(x) = [t^m]g(x, t)$ . On  $t = e^{i\theta}$ , the exponential becomes  $e^{ix \sin \theta}$ : this explains its connection to waves in circular geometry. Recall Laurent coefficient extraction from Chapter 1:

$$[t^m]f(t) = \frac{1}{2\pi i} \oint_C f(t) t^{-m-1} dt, \quad (5.1.2)$$

where  $C$  is a counterclockwise circle about zero inside the annulus of convergence. Only the term proportional to  $t^{-1}$  survives the integration.

We will multiply the two exponential series to recover the Chapter 3 Bessel series, then differentiate the same generating function to relate neighboring orders.

# Reading off the integer-order series

**Expand, then collect one power.** Since  $g = e^{xt/2} e^{-x/(2t)}$ ,

$$e^{xt/2} = \sum_{j=0}^{\infty} \frac{(x/2)^j}{j!} t^j, \quad e^{-x/(2t)} = \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^k}{k!} t^{-k}.$$

Their product is

$$g(x, t) = \sum_{j,k \geq 0} \frac{(-1)^k}{j!k!} \left(\frac{x}{2}\right)^{j+k} t^{j-k}.$$

For  $t \neq 0$  both exponential series converge absolutely, so we may group terms. To obtain  $t^m$  with  $m \geq 0$ , require

$$j - k = m \implies j = k + m, \quad j + k = 2k + m.$$

Substitute this into the coefficient of the double sum:

$$J_m(x) = [t^m]g(x, t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m}, \quad m = 0, 1, 2, \dots \quad (5.1.3)$$

## Reading off the integer-order series (cont.)

For example,

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \cdots,$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} + \cdots,$$

$$J_2(x) = \frac{x^2}{8} - \frac{x^4}{96} + \frac{x^6}{3072} + \cdots.$$

These are the series obtained from the differential equation in Chapter 3. A single generating function now holds all the integer orders at once. The general-order function  $J_\nu$  below is not a Laurent coefficient when  $\nu \notin \mathbb{Z}$ .

# General order series

Replacing factorials by the gamma function extends the series to general order:

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}. \quad (5.1.4)$$

At a nonpositive integer,  $1/\Gamma$  is defined by continuity and equals zero. Use this formula directly at negative integer order; do not divide a recurrence by zero. For  $x > 0$ ,  $(x/2)^\nu = e^{\nu \log(x/2)}$  is unambiguous; complex  $x$  requires a fixed logarithm branch when  $\nu$  is noninteger.

For fixed order, the ratio of consecutive nonzero terms eventually has magnitude  $|x|^2/[4(k+1)(k+\nu+1)] \rightarrow 0$ . The power-series factor converges for every finite  $x$ . For  $\nu = m \geq 0$ ,  $\Gamma(k+m+1) = (k+m)!$ , recovering (5.1.3).

## Negative integer orders: $J_{-m} = (-1)^m J_m$

In the double sum for  $g$ , a coefficient of  $t^{-m}$  requires  $j - k = -m$ . For  $m \geq 0$ , this means  $k = j + m$ . Thus

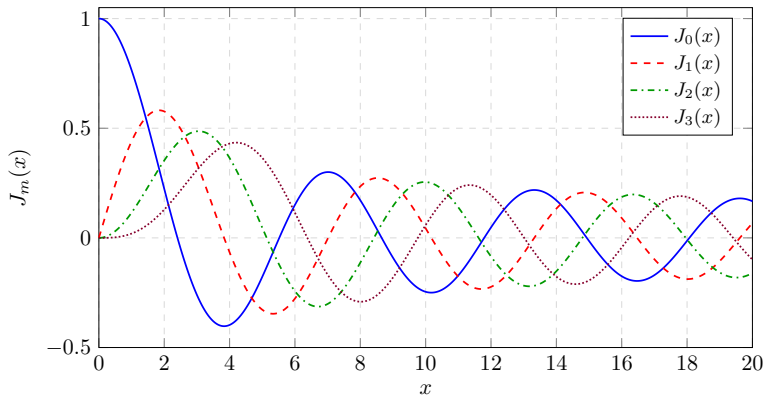
$$\begin{aligned} J_{-m}(x) &= \sum_{j=0}^{\infty} \frac{(-1)^{j+m}}{j!(j+m)!} \left(\frac{x}{2}\right)^{2j+m} \\ &= (-1)^m \sum_{j=0}^{\infty} \frac{(-1)^j}{j!(j+m)!} \left(\frac{x}{2}\right)^{2j+m}. \end{aligned}$$

The remaining sum is  $J_m$ , so

$$J_{-m}(x) = (-1)^m J_m(x), \quad m \in \mathbb{Z}. \quad (5.1.5)$$

The gamma series agrees: its first  $m$  coefficients at order  $-m$  vanish because  $1/\Gamma$  is zero at nonpositive integers. In particular,  $J_{-1} = -J_1$  and  $J_{-2} = J_2$ . Negative integer orders therefore do not provide a second independent solution.

# Bessel functions of the first kind



For  $m \geq 0$ , the leading term in (5.1.3) is  $J_m(x) = (x/2)^m/m! + O(x^{m+2})$ . Thus  $J_0(0) = 1$  and  $J_m(0) = 0$  for  $m \geq 1$ .

# Jacobi–Anger: the generating function on $|t| = 1$

Set  $t = e^{i\theta}$  in (5.1.1). Since  $t - t^{-1} = 2i \sin \theta$  and  $t^m = e^{im\theta}$ ,

$$e^{ix \sin \theta} = \sum_{m=-\infty}^{\infty} J_m(x) e^{im\theta}. \quad (5.1.6)$$

This is the Jacobi–Anger expansion: the Bessel functions are Fourier coefficients of a plane wave written in angular variables.

To isolate  $J_n$ , multiply by  $e^{-in\theta}$  and integrate over one period. For an integer  $q$ ,

$$\int_0^{2\pi} e^{iq\theta} d\theta = \begin{cases} 2\pi, & q = 0, \\ (e^{2\pi iq} - 1)/(iq) = 0, & q \neq 0. \end{cases}$$

The series may be integrated term by term on this circle. All terms with  $m \neq n$  vanish, leaving

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z}. \quad (5.1.7)$$

We will convert this to a real cosine integral after deriving the recurrences.

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# Differentiating in $t$ : the three-term recurrence

Differentiate  $g = e^{x(t-t^{-1})/2}$  and multiply by  $t$  before comparing coefficients:

$$tg_t = \frac{x}{2}(t + t^{-1})g, \quad tg_t = \sum_{m=-\infty}^{\infty} mJ_m(x)t^m. \quad (5.2.1)$$

Multiplication by  $t$  moves the coefficient of  $t^{m-1}$  to  $t^m$ ; multiplication by  $t^{-1}$  moves the coefficient of  $t^{m+1}$  to  $t^m$ . Consequently,

$$[t^m](tg) = J_{m-1}(x), \quad [t^m](t^{-1}g) = J_{m+1}(x).$$

Matching the coefficient of  $t^m$  in (5.2.1) gives

$$mJ_m(x) = \frac{x}{2}[J_{m-1}(x) + J_{m+1}(x)].$$

For  $x \neq 0$ , divide by  $x/2$ :

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x}J_m(x). \quad (5.2.2)$$

At  $x = 0$ , use limits from the power series, not direct substitution into a quotient.

# Differentiating in $x$ : the derivative recurrence

Differentiate the same generating function in  $x$ :

$$g_x = \frac{1}{2}(t - t^{-1})g, \quad g_x = \sum_{m=-\infty}^{\infty} J'_m(x)t^m. \quad (5.2.3)$$

The coefficient shifts are the same as before, with a minus sign between them:

$$2J'_m(x) = J_{m-1}(x) - J_{m+1}(x). \quad (5.2.4)$$

Now add and subtract (5.2.2) and (5.2.4):

$$2J_{m-1} = \frac{2m}{x}J_m + 2J'_m, \quad 2J_{m+1} = \frac{2m}{x}J_m - 2J'_m.$$

Solving each for  $J'_m$  gives, for  $x \neq 0$ ,

$$J'_m = J_{m-1} - \frac{m}{x}J_m, \quad (5.2.5)$$

$$J'_m = \frac{m}{x}J_m - J_{m+1}. \quad (5.2.6)$$

These are exact symbolic identities. Repeated forward use at fixed  $x$  and large  $m$  can amplify roundoff; an identity need not be a stable numerical algorithm.

# Consequences and low-order checks

Putting  $m = 0$  in (5.2.6) gives

$$J_0'(x) = -J_1(x). \quad (5.2.7)$$

One series check fixes the sign:

$$\frac{d}{dx} \left( 1 - \frac{x^2}{4} + \frac{x^4}{64} + \cdots \right) = -\frac{x}{2} + \frac{x^3}{16} + \cdots = -J_1(x).$$

The one-sided recurrences also turn products into derivatives. For  $x \neq 0$ ,

$$\frac{d}{dx}(x^m J_m) = x^m \left( J_m' + \frac{m}{x} J_m \right) = x^m J_{m-1}, \quad (5.2.8)$$

$$\frac{d}{dx}(x^{-m} J_m) = x^{-m} \left( J_m' - \frac{m}{x} J_m \right) = -x^{-m} J_{m+1}. \quad (5.2.9)$$

The first equality on each line is the product rule; the second uses the corresponding one-sided derivative recurrence. These forms are useful in integrals and boundary conditions.

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# Bessel's real integral representation

For real  $x$ , start from the Jacobi–Anger expansion (5.1.6) and pair the orders  $m$  and  $-m$ . Since  $J_{-m} = (-1)^m J_m$ , the even orders combine into cosines and the odd orders into sines:

$$\begin{aligned} e^{ix \sin \theta} &= J_0(x) + 2 \sum_{k=1}^{\infty} J_{2k}(x) \cos(2k\theta) \\ &\quad + 2i \sum_{k=0}^{\infty} J_{2k+1}(x) \sin((2k+1)\theta). \end{aligned}$$

Taking real and imaginary parts gives

$$\begin{aligned} \cos(x \sin \theta) &= J_0(x) + 2 \sum_{k=1}^{\infty} J_{2k}(x) \cos(2k\theta), \\ \sin(x \sin \theta) &= 2 \sum_{k=0}^{\infty} J_{2k+1}(x) \sin((2k+1)\theta). \end{aligned}$$

# Bessel's real integral representation (cont.)

Project these two Fourier series on  $[0, \pi]$ . The orthogonality integrals are

$$\int_0^\pi \cos(j\theta) \cos(m\theta) d\theta = \int_0^\pi \sin(j\theta) \sin(m\theta) d\theta = \frac{\pi}{2} \delta_{jm}$$

for positive  $j, m$ ; the constant mode has  $\int_0^\pi 1 d\theta = \pi$ . Hence, with the parity conditions correctly attached to  $m$  and with the factor  $\pi$  retained,

$$\begin{aligned} \int_0^\pi \cos(x \sin \theta) \cos(m\theta) d\theta &= \begin{cases} \pi J_m(x), & m \text{ even,} \\ 0, & m \text{ odd,} \end{cases} \\ \int_0^\pi \sin(x \sin \theta) \sin(m\theta) d\theta &= \begin{cases} 0, & m \text{ even,} \\ \pi J_m(x), & m \text{ odd.} \end{cases} \end{aligned}$$

The first formula includes  $m = 0$ . Adding the two projections and using  $\cos(A - B) = \cos A \cos B + \sin A \sin B$  gives

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta, \quad m \in \mathbb{Z}_{\geq 0}. \quad (5.3.1)$$

# Bessel's real integral representation (cont.)

In particular,

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) d\theta. \quad (5.3.2)$$

Negative orders then follow from (5.1.5).

# Contour form (Schl\"afli / Cauchy)

Laurent extraction (5.1.2) applied to the generating function gives

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}, \quad (5.3.3)$$

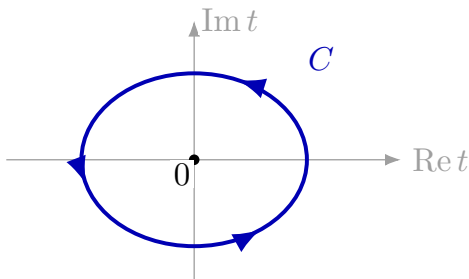
where  $C$  is a simple counterclockwise contour encircling zero once, without passing through zero. On the unit circle  $t = e^{i\theta}$ ,

$$dt = it d\theta, \quad \frac{x}{2}(t - t^{-1}) = ix \sin \theta, \quad t^{-n-1} dt = ie^{-in\theta} d\theta.$$

Substitution recovers (5.1.7). This is the Chapter 1 coefficient formula applied to the whole Bessel family.

# Contours in the Schlöfli representation

integer order



$t^{-n-1}$  single-valued  
one counterclockwise circuit

For integer  $n$ ,  $t^{-n-1}$  is single-valued. The contour travels counterclockwise once around zero; its shape can change without changing the coefficient integral.

# Noninteger order needs an open contour

The Schlöfli loop (5.3.3) is legal only because  $t^{-n-1}$  is single-valued for integer  $n$ . For a general order  $\nu$ , the power  $t^{-\nu-1} = e^{-(\nu+1)\log t}$  needs a branch: take  $-\pi < \arg t < \pi$  with the cut on the negative real axis. A closed loop can no longer cross the cut, so it opens into the Hankel contour  $\mathcal{H}$  of Chapter 2: from  $-\infty$  below the cut, counterclockwise around the origin, and back to  $-\infty$  above. Define

$$\mathcal{J}_\nu(x) = \frac{1}{2\pi i} \int_{\mathcal{H}} e^{\frac{x}{2}(t-t^{-1})} t^{-\nu-1} dt, \quad x > 0. \quad (5.3.4)$$

Both legs converge, since  $e^{xt/2} \rightarrow 0$  as  $t \rightarrow -\infty$ .

**Which Bessel function is this?** Expand only the descending exponential,  $e^{-x/(2t)} = \sum_q (-x/2t)^q / q!$ ; on the truncated contour the series converges uniformly, so termwise integration followed by the decaying-leg limit is justified. In the  $q$ th term set  $u = xt/2$ : the contour is  $\mathcal{H}$  scaled by  $x/2$ , and  $t^{-\nu-q-1} dt = (x/2)^{\nu+q} u^{-\nu-q-1} du$ .

Hankel's reciprocal formula for  $1/\Gamma$  from Chapter 2 gives

$$\frac{1}{2\pi i} \int_{\mathcal{H}} e^{xt/2} t^{-\nu-q-1} dt = \frac{(x/2)^{\nu+q}}{\Gamma(\nu+q+1)},$$

# Noninteger order needs an open contour (cont.)

so (5.3.4) is term-by-term the general-order series (5.1.4):

$$\mathcal{J}_\nu(x) = \sum_{q=0}^{\infty} \frac{(-1)^q (x/2)^{\nu+2q}}{q! \Gamma(\nu + q + 1)} = J_\nu(x). \quad (5.3.5)$$

This identification is the Sommerfeld–Watson reduction: an open contour plus the  $1/\Gamma$  identity replaces the integer-order coefficient extraction.

# Sommerfeld's form on the $z$ -plane

Put  $t = ie^{-iz} = e^{i(\pi/2-z)}$  in (5.3.4). The pieces are

$$\frac{x}{2}(t - t^{-1}) = ix \cos z, \quad t^{-\nu} = e^{i\nu(z-\pi/2)}, \quad \frac{dt}{t} = -i dz.$$

The Hankel contour  $\mathcal{H}$  maps to *Sommerfeld's contour*  $C$ , which runs from  $z = -\pi/2 + i\infty$  down to the real axis and back up to  $z = 3\pi/2 + i\infty$ : at both ends  $\cos z = i \sinh y$  for  $z = -\pi/2 + iy$  or  $3\pi/2 + iy$ , so  $e^{ix \cos z} = e^{-x \sinh y}$  decays doubly-exponentially and the open ends are legal for  $x > 0$ . Assembling the pieces,  $t^{-\nu-1} dt = t^{-\nu} (dt/t) = -i e^{i\nu(z-\pi/2)} dz$ , so the integrand of (5.3.4) is  $-i e^{i[x \cos z + \nu(z-\pi/2)]} dz$  and the prefactor becomes  $\frac{1}{2\pi i}(-i) = -\frac{1}{2\pi}$ . The substitution also reverses the orientation of the contour, and that reversal supplies a second minus sign, cancelling the first and leaving

$$J_\nu(x) = \frac{1}{2\pi} \int_C e^{i[x \cos z + \nu(z-\pi/2)]} dz, \quad (5.3.6)$$

the form used when a field is written as a wave-function integral in the angle variable  $z$ .

**Why the detour.** Representation (5.3.6) keeps working when  $\nu$  is not an integer—the price is the open contour, not a new function. When an angular spectral sum is wrapped into a  $\nu$ -plane integral (as in cylinder and wedge problems), (5.3.6) is what identifies the integrand as  $J_\nu$  and lets its poles and saddles be read off directly.

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## Small $x$ : leading term and first corrections

For fixed  $m \geq 0$ , take the first three terms of (5.1.3):

$$J_m(x) = \frac{(x/2)^m}{m!} - \frac{(x/2)^{m+2}}{(m+1)!} + \frac{(x/2)^{m+4}}{2!(m+2)!} + O(x^{m+6}).$$

Factor out  $(x/2)^m/m!$ . Using  $(m+1)! = (m+1)m!$  and  $(m+2)! = (m+2)(m+1)m!$  gives

$$J_m(x) = \frac{(x/2)^m}{m!} \left[ 1 - \frac{x^2}{4(m+1)} + \frac{x^4}{32(m+1)(m+2)} + O(x^6) \right]. \quad (5.4.1)$$

In particular,

$$J_m(x) = \frac{(x/2)^m}{\Gamma(m+1)} + O(x^{m+2}), \quad x \rightarrow 0, \quad (5.4.2)$$

with the order held fixed. The gamma notation agrees with the general-order series introduced earlier. Thus  $J_0(0) = 1$  and  $J_m(0) = 0$  for  $m \geq 1$ .

# Large $x$ : steepest descent on the Schlöfli contour

Apply steepest descent directly to the contour representation. For fixed integer  $n$ , write (5.3.3) as

$$J_n(x) = \frac{1}{2\pi i} \oint_C g(t) e^{xf(t)} dt, \quad f(t) = \frac{1}{2}(t - t^{-1}), \quad g(t) = t^{-n-1}.$$

Here  $x \rightarrow +\infty$  is the large parameter. The saddle points satisfy

$$f'(t) = \frac{1}{2}(1 + t^{-2}) = 0, \quad t_+ = i, \quad t_- = -i,$$

and

$$f''(t) = -t^{-3}.$$

The values needed in the local approximations are

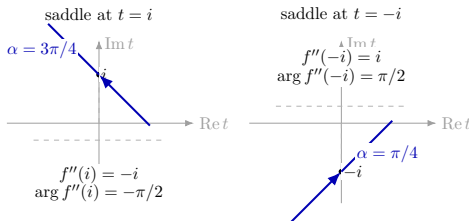
| $t_0$ | $f(t_0)$ | $f''(t_0)$ | $g(t_0)$           |
|-------|----------|------------|--------------------|
| $i$   | $i$      | $-i$       | $e^{-i(n+1)\pi/2}$ |
| $-i$  | $-i$     | $i$        | $e^{i(n+1)\pi/2}$  |

# Large $x$ : steepest descent on the Schlöfli contour (cont.)

If  $t = t_0 + e^{i\alpha} u$  with real  $u$ , the quadratic exponential is

$$\exp \left[ \frac{x}{2} f''(t_0) e^{2i\alpha} u^2 \right].$$

It decays most rapidly when  $f''(t_0)e^{2i\alpha}$  is negative real. Thus the steepest lines have angle  $3\pi/4$  through  $i$  and angle  $\pi/4$  through  $-i$ . The original counterclockwise contour can be deformed through both saddles with these orientations and without crossing the singularity at the origin.



# Large $x$ : combining the two saddle contributions

**Contribution from  $t = i$ .** Put  $t = i + e^{3\pi i/4}u$ . Then

$$f(t) = i - \frac{1}{2}u^2 + O(u^3), \quad g(t) = e^{-i(n+1)\pi/2} [1 + O(u)], \quad dt = e^{3\pi i/4} du.$$

Keeping the quadratic term and constant amplitude gives the contribution to the contour integral before the factor  $1/(2\pi i)$ :

$$\begin{aligned} I_i &\sim e^{ix} e^{-i(n+1)\pi/2} e^{3\pi i/4} \int_{-\infty}^{\infty} e^{-xu^2/2} du \\ &= i \sqrt{\frac{2\pi}{x}} e^{i(x - n\pi/2 - \pi/4)}. \end{aligned}$$

Here  $\int_{-\infty}^{\infty} e^{-xu^2/2} du = \sqrt{2\pi/x}$ .

**Contribution from  $t = -i$ .** With  $t = -i + e^{\pi i/4}u$ ,

$$f(t) = -i - \frac{1}{2}u^2 + O(u^3), \quad g(t) = e^{i(n+1)\pi/2} [1 + O(u)], \quad dt = e^{\pi i/4} du,$$

so

$$I_{-i} \sim i \sqrt{\frac{2\pi}{x}} e^{-i(x - n\pi/2 - \pi/4)}.$$

## Large $x$ : combining the two saddle contributions (cont.)

Set  $\chi_n(x) = x - n\pi/2 - \pi/4$ . Adding both saddle contributions and restoring the prefactor gives

$$J_n(x) = \frac{I_i + I_{-i}}{2\pi i} + O(x^{-3/2}) = \frac{e^{i\chi_n(x)} + e^{-i\chi_n(x)}}{\sqrt{2\pi x}} + O(x^{-3/2}).$$

Since  $e^{i\chi} + e^{-i\chi} = 2 \cos \chi$ ,

$$J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}), \quad x \rightarrow +\infty. \quad (5.4.3)$$

This holds for fixed integer  $n$ . We use an absolute error rather than a ratio asymptotic because the leading cosine has infinitely many zeros.

For fixed noninteger real order the same formula holds with  $n$  replaced by  $\nu$  (quoted; DLMF §10.17). The integer Fourier integral does not derive that extension. In both cases the order is held fixed as  $x \rightarrow +\infty$ .

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# Reduction of order: constructing a second solution

Dividing Bessel's equation by  $x^2$  for  $x > 0$  gives

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0. \quad (5.5.1)$$

Chapter 3, §3.3 derived the reduction-of-order formula  $y_2 = y_1 \int e^{-\int P dx} / y_1^2 dx$  for  $y'' + Py' + Qy = 0$ . Here  $P = 1/x$  and  $e^{-\int P dx} = 1/x$ , so

$$y_2(x) = J_m(x) \int_{x_*}^x \frac{ds}{sJ_m^2(s)}.$$

This formula is valid on an interval containing  $x_*$  where  $J_m$  has no zero. It supplies the general solution there:

$$y(x) = AJ_m(x) + BJ_m(x) \int_{x_*}^x \frac{ds}{sJ_m^2(s)}. \quad (5.5.2)$$

# Reduction of order: constructing a second solution (cont.)

For  $v = \int_{x_*}^x ds/[sJ_m^2(s)]$ , the product-rule terms cancel:

$$W[J_m, vJ_m] = J_m(v'J_m + vJ_m') - J_m'(vJ_m) = v'J_m^2 = \frac{1}{x}.$$

The conventional Neumann function has normalization

$$W[J_m, Y_m] = \frac{2}{\pi x}. \quad (5.5.3)$$

Thus  $y_2 = (\pi/2)Y_m + cJ_m$ . Multiplying by  $2/\pi$  fixes the Wronskian, but not the added multiple of  $J_m$ . The definitions below fix that remaining choice. The integral formula is local; the resulting solution extends across positive zeros of  $J_m$ .

# Noninteger order: definition and independence of $Y_\nu$

Bessel's equation depends on the order only through  $\nu^2$ . Therefore, if  $J_\nu(x)$  is a solution, then  $J_{-\nu}(x)$  is also a solution. For fixed real  $\nu \notin \mathbb{Z}$ , define

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}. \quad (5.5.4)$$

The two solutions are independent, with the standard normalization

$$W[J_\nu, Y_\nu] = \frac{2}{\pi x}.$$

Abel's identity fixes the  $1/x$  dependence; the constant follows from the small- $x$  leading terms and Euler's reflection formula.

At integer order the quotient has a removable  $0/0$  form. Its continuous limit defines  $Y_m$ ; the next frame states that definition. The Wronskian also passes to the limit and remains  $2/(\pi x)$ , so independence is retained.

## Integer order: define the second solution by a limit

At an integer  $m$ ,  $J_{-m} = (-1)^m J_m$ ,  $\cos(\pi m) = (-1)^m$ , and  $\sin(\pi m) = 0$ . Thus direct substitution into (5.5.4) gives  $0/0$ , not a value for  $Y_m$ .

The conventional definition is the continuous limit

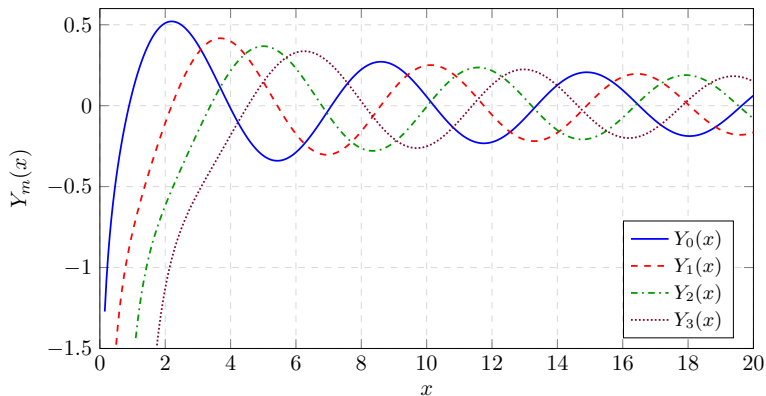
$$Y_m(x) = \lim_{\nu \rightarrow m} \frac{J_\nu(x) \cos(\pi \nu) - J_{-\nu}(x)}{\sin(\pi \nu)}, \quad x > 0. \quad (5.5.5)$$

We use the standard result that this limit, and its first two  $x$ -derivatives, exist. It therefore still solves the integer-order Bessel equation. The Wronskian passes to the limit:

$$W[J_m, Y_m] = \frac{2}{\pi x} \neq 0.$$

So  $Y_m$  is independent of  $J_m$ , even though  $J_{-m}$  is not. Unlike a Wronskian normalization alone, this limit also fixes the additive multiple of  $J_m$ . We use  $Y_m$  directly; order derivatives are not needed below.

# Bessel functions of the second kind



The Neumann functions are independent of  $J_m$  and singular at  $x = 0$ . For large positive  $x$ ,  $J_m$  and  $Y_m$  have the same  $x^{-1/2}$  envelope and phases shifted by  $\pi/2$ .

## Small- $x$ behavior of $Y_m$

First let  $\nu > 0$  be fixed and noninteger. From the first term of the series,

$$J_\nu(x) = \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu [1 + O(x^2)],$$
$$J_{-\nu}(x) = \frac{1}{\Gamma(1-\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + O(x^2)].$$

Their magnitudes have orders  $x^\nu$  and  $x^{-\nu}$ , respectively. Since  $x^\nu/x^{-\nu} = x^{2\nu} \rightarrow 0$ , the  $J_{-\nu}$  term dominates in (5.5.4). Therefore

$$Y_\nu(x) = -\frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + o(1)].$$

Euler's reflection formula gives

$$\Gamma(\nu)\Gamma(1-\nu) = \frac{\pi}{\sin(\pi\nu)} \quad \implies \quad \frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} = \frac{\Gamma(\nu)}{\pi}.$$

Consequently,

$$Y_\nu(x) = -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu [1 + o(1)], \quad x \rightarrow 0^+, \quad (5.5.6)$$

for fixed positive noninteger  $\nu$ .

## Small- $x$ behavior of $Y_m$ (cont.)

For positive integer  $m$ , the reduction of order in Chapter 3 gives the same leading behavior, so

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad m = 1, 2, \dots, \quad x \rightarrow 0^+. \quad (5.5.7)$$

For  $m = 0$ ,  $J_0(x) = 1 + O(x^2)$ , so the integrand is  $1/[xJ_0^2(x)] = 1/x + O(x)$ . Its integral is  $\ln x + C + O(x^2)$ . The order-limit definition fixes the constant and gives the standard formula

$$Y_0(x) = \frac{2}{\pi} \left[ \ln \left( \frac{x}{2} \right) + \gamma \right] + O(x^2 |\ln x|), \quad x \rightarrow 0^+. \quad (5.5.8)$$

The leading logarithm follows from  $\int dx/x$ ; the additive constant is quoted from the standard normalization (DLMF §10.8), not determined by that leading integral.

# Large- $x$ behavior of $J_\nu$ and $Y_\nu$

For fixed real order  $\nu$ , set

$$\chi_\nu(x) = x - \frac{\nu\pi}{2} - \frac{\pi}{4}, \quad A(x) = \sqrt{\frac{2}{\pi x}}.$$

The preceding large- $x$  result and its quoted noninteger extension give

$$J_\nu(x) = A(x) \cos \chi_\nu(x) + O(x^{-3/2}). \quad (5.5.9)$$

Since  $\chi_{-\nu} = \chi_\nu + \pi\nu$ , insert the leading terms into the definition of  $Y_\nu$  for fixed noninteger  $\nu$ :

$$Y_\nu(x) = A(x) \frac{\cos \chi_\nu \cos(\pi\nu) - \cos(\chi_\nu + \pi\nu)}{\sin(\pi\nu)} + O(x^{-3/2}).$$

The identity  $\cos(\chi + \beta) = \cos \chi \cos \beta - \sin \chi \sin \beta$  makes the numerator  $\sin \chi_\nu \sin(\pi\nu)$ , so

$$Y_\nu(x) = \sqrt{\frac{2}{\pi x}} \sin \chi_\nu(x) + O(x^{-3/2}), \quad x \rightarrow +\infty. \quad (5.5.10)$$

The same sine law holds at integer order (quoted; DLMF §10.17); the division above is not a proof of that limit. Thus  $J_m$  and  $Y_m$  have the same leading envelope and a phase difference of  $\pi/2$ .

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# Hankel functions: Bessel functions of the third kind

Define

$$H_m^{(1)}(x) = J_m(x) + iY_m(x), \quad (5.6.1)$$

$$H_m^{(2)}(x) = J_m(x) - iY_m(x). \quad (5.6.2)$$

Let

$$\chi_m(x) = x - \frac{m\pi}{2} - \frac{\pi}{4}.$$

Using (5.5.9) and (5.5.10),

$$\begin{aligned} H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m + i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \end{aligned}$$

$$\begin{aligned} H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m - i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}). \end{aligned}$$

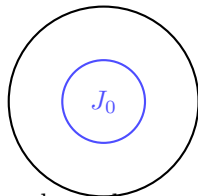
# Hankel functions: Bessel functions of the third kind (cont.)

With  $k > 0$ ,  $\omega > 0$ , and time dependence  $e^{-i\omega t}$ ,

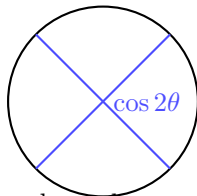
$$H_m^{(1)}(kr)e^{-i\omega t} \sim \sqrt{\frac{2}{\pi kr}} e^{i(kr - \omega t - m\pi/2 - \pi/4)}, \quad r \rightarrow \infty$$

for fixed  $m$ . Its constant-phase surfaces move toward larger  $r$ , so it is outgoing;  $H_m^{(2)}$  is incoming. The Hankel functions are exact radial solutions for  $r > 0$ ; only these leading exponential forms are far-field approximations. Both are singular at the origin, unlike the regular  $J_m$  modes.

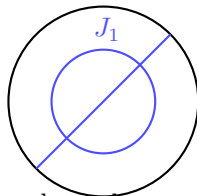
# Discrete radial modes on the unit disk



angular order  $m = 0$   
radial mode 2



angular order  $m = 2$   
radial mode 1  
 $\theta$  is the polar angle



angular order  $m = 1$   
radial mode 2

The boundary condition  $J_m(\alpha_{m,j}) = 0$  places a radial zero at the clamped rim  $r = 1$ . Distinct zeros give distinct Sturm–Liouville eigenvalues  $\lambda_j = \alpha_{m,j}^2$ .

# Scaling Bessel's equation and obtaining Sturm–Liouville form

Return to the disk problem in Chapter 4, §4.2. For fixed integer  $m \geq 0$  and  $y(s) = J_m(\alpha s)$ , the chain rule gives  $y' = \alpha J'_m$  and  $y'' = \alpha^2 J''_m$ . The rescaled equation is

$$s^2 y'' + sy' + (\alpha^2 s^2 - m^2)y = 0. \quad (5.6.3)$$

Divide by  $s > 0$  and use  $(sy')' = sy'' + y'$ :

$$(sy')' - \frac{m^2}{s}y = -\alpha^2 sy. \quad (5.6.4)$$

Thus  $p = s$ ,  $q = -m^2/s$ ,  $w = s$ , and  $\lambda = \alpha^2$ . The weight gives

$$\langle u, v \rangle = \int_0^1 u(s) \overline{v(s)} s \, ds, \quad (5.6.5)$$

linear in the first slot, as in Chapter 4. The disk modes are regular at the origin and satisfy  $y(1) = 0$  at the rim.

# Scaling Bessel's equation and obtaining Sturm–Liouville form (cont.)

**Check both endpoints.** Green's identity leaves the boundary expression  $s(uv' - u'v)$ . For two regular Bessel solutions,

|            | $u, v$       | $u', v'$     | $s(uv' - u'v)$ |
|------------|--------------|--------------|----------------|
| $m \geq 1$ | $O(s^m)$     | $O(s^{m-1})$ | $O(s^{2m})$    |
| $m = 0$    | $1 + O(s^2)$ | $O(s)$       | $O(s^2)$       |

so the expression tends to zero at the origin in both cases. Singular  $Y_m$  modes are excluded by the regularity condition. At the rim both mode values vanish. This is why Chapter 4's orthogonality result applies;  $p(0) = 0$  alone would not suffice. The remaining task is new: compute the diagonal norm needed for expansion coefficients.

# Fourier–Bessel orthogonality: distinct eigenvalues

For positive parameters  $a, b$ , let  $y_a(x) = J_m(ax)$  and  $y_b(x) = J_m(bx)$ . Chapter 4's Green identity applied to (5.6.4) gives

$$\frac{d}{dx} [x(y_b y'_a - y_a y'_b)] + (a^2 - b^2)xy_a y_b = 0.$$

Integrating and using the origin estimate just checked,

$$(a^2 - b^2) \int_0^1 xy_a y_b dx = -[x(y_b y'_a - y_a y'_b)]_0^1.$$

At  $x = 1$ ,  $y_a = J_m(a)$  and  $y'_a = aJ'_m(a)$ ; similarly for  $b$ . Hence

$$\int_0^1 xJ_m(ax)J_m(bx) dx = \frac{bJ_m(a)J'_m(b) - aJ_m(b)J'_m(a)}{a^2 - b^2}, \quad a \neq b. \quad (5.6.6)$$

The order of the two numerator terms comes from the minus sign before the boundary term.

## Fourier–Bessel orthogonality: distinct eigenvalues (cont.)

Now choose distinct positive zeros  $a = \alpha_{m,j}$  and  $b = \alpha_{m,k}$  of  $J_m$ . Both  $J_m(a)$  and  $J_m(b)$  are zero, so the numerator vanishes while  $a^2 - b^2 \neq 0$ :

$$\int_0^1 x J_m(\alpha_{m,j} x) J_m(\alpha_{m,k} x) dx = 0, \quad j \neq k. \quad (5.6.7)$$

For the diagonal case  $j = k$ , both numerator and denominator vanish instead. We therefore take a limit, rather than substitute  $a = b$  into (5.6.6).

# Fourier–Bessel normalization: the diagonal case

**Existence of the zeros.** For fixed integer  $m \geq 0$ , the large- $x$  cosine law (5.4.3) oscillates with amplitude  $\sim x^{-1/2}$  and phase advancing like  $x$ , so  $J_m$  changes sign infinitely often as  $x \rightarrow +\infty$ . Hence  $J_m$  has infinitely many positive zeros

$\alpha_{m,1} < \alpha_{m,2} < \cdots$ . We write  $\alpha_{m,j}$  for the  $j$ th.

Every positive zero  $a$  is simple: at  $J_m(a) = 0$ , the Wronskian gives

$-J'_m(a)Y_m(a) = 2/(\pi a) \neq 0$ , so  $J'_m(a) \neq 0$ . This ensures that the norm in the coefficient formula is nonzero.

To compute the norm at such a zero, take the limit  $b \rightarrow a$  in (5.6.6). The integral passes to the limit by continuity of the integrand on the compact interval  $0 \leq x \leq 1$ .

Write the numerator and denominator as functions of  $b$  with  $a$  fixed and  $J_m(a) = 0$ :

$$N(b) := bJ_m(a)J'_m(b) - aJ_m(b)J'_m(a) = -aJ_m(b)J'_m(a),$$

$$D(b) := a^2 - b^2.$$

## Fourier–Bessel normalization: the diagonal case (cont.)

Both  $N(a)$  and  $D(a)$  vanish, so the limit is a  $0/0$  form. Apply L'Hôpital: differentiate with respect to  $b$ ,

$$\begin{aligned}N'(b) &= -aJ'_m(b)J'_m(a), \\D'(b) &= -2b.\end{aligned}$$

Therefore

$$\lim_{b \rightarrow a} \frac{N(b)}{D(b)} = \lim_{b \rightarrow a} \frac{N'(b)}{D'(b)} = \frac{-aJ'_m(a)J'_m(a)}{-2a} = \frac{1}{2} [J'_m(a)]^2,$$

where  $a > 0$ . Therefore

$$\int_0^1 x J_m^2(ax) dx = \frac{1}{2} [J'_m(a)]^2 \quad \text{at a positive zero } a \text{ of } J_m. \quad (5.6.8)$$

At a zero,  $J_m(a) = 0$ , so the derivative recurrence gives

$$J'_m(a) = \frac{m}{a} J_m(a) - J_{m+1}(a) = -J_{m+1}(a).$$

Squaring removes the sign. Combining the diagonal and off-diagonal cases gives

$$\int_0^1 J_m(\alpha_{m,j}x) J_m(\alpha_{m,k}x) x dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk}. \quad (5.6.9)$$

# Fourier–Bessel coefficients

For a radial function of fixed angular order  $m$ , write

$$f(x) = \sum_{j=1}^{\infty} a_j J_m(\alpha_{m,j}x), \quad 0 < x < 1. \quad (5.6.10)$$

As for the sine series in Chapter 4, equality means convergence in the weighted mean-square norm  $\|g\|^2 = \int_0^1 |g(x)|^2 x \, dx$ .

Apply the same projection formula: multiply by the  $k$ th mode and the weight  $x$ , then integrate. Orthogonality removes every term except  $j = k$ :

$$\int_0^1 f(x) J_m(\alpha_{m,k}x) x \, dx = a_k \int_0^1 J_m^2(\alpha_{m,k}x) x \, dx = \frac{a_k}{2} J_{m+1}^2(\alpha_{m,k}).$$

Solving gives

$$a_k = \frac{2}{J_{m+1}^2(\alpha_{m,k})} \int_0^1 f(r) J_m(\alpha_{m,k}r) r \, dr. \quad (5.6.11)$$

The denominator is the norm just derived, not an arbitrary normalization.

These disk modes are complete in the weighted mean-square space (a quoted Sturm–Liouville result). Finite weighted norm therefore suffices for the expansion, but not for pointwise equality.

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# The three-dimensional radial equation

Separating the Helmholtz equation  $(\nabla^2 + k^2)\psi = 0$  in spherical coordinates produces the angular eigenvalue  $\lambda = n(n+1)$  already met in Laplace's equation (Chapter 6) and the radial equation

$$x^2 R'' + 2xR' + [x^2 - n(n+1)]R = 0, \quad x = kr.$$

With  $u = xR$  the first derivative drops out, since  $x^2 R'' + 2xR' = xu''$  for  $R = u/x$ :

$$u'' + \left[1 - \frac{n(n+1)}{x^2}\right]u = 0. \quad (5.7.1)$$

On the other hand, substitute  $J_{n+1/2} = x^{-1/2}v$  (so  $v = \sqrt{x} J_{n+1/2}$ ) into Bessel's equation of order  $\nu = n + \frac{1}{2}$ . This is the standard first-derivative-removing substitution  $y = v \exp(-\frac{1}{2} \int P dx) = x^{-1/2}v$  with  $P = 1/x$ ; it cancels the  $y'$  term and shifts the potential by  $-\frac{1}{4}P^2 - \frac{1}{2}P' = +\frac{1}{4x^2}$ , turning the  $-\nu^2/x^2$  coefficient into  $-(\nu^2 - \frac{1}{4})/x^2$ . Hence

$$v'' + \left[1 - \frac{(n + \frac{1}{2})^2 - \frac{1}{4}}{x^2}\right]v = 0,$$

# The three-dimensional radial equation (cont.)

and  $(n + \frac{1}{2})^2 - \frac{1}{4} = n(n+1)$ : the equations coincide. The regular and singular radial solutions are therefore the *spherical Bessel functions*

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+1/2}(x), \quad y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+1/2}(x), \quad (5.7.2)$$

and the outgoing/incoming pairs are the *spherical Hankel functions*

$$h_n^{(1)} = j_n + iy_n = \sqrt{\frac{\pi}{2x}} H_{n+1/2}^{(1)}, \quad h_n^{(2)} = j_n - iy_n = \sqrt{\frac{\pi}{2x}} H_{n+1/2}^{(2)}. \quad (5.7.3)$$

The Riccati–Bessel functions  $\hat{z}_n(x) = xz_n(x)$  ( $z = j, y, h^{(1)}, h^{(2)}$ ) satisfy (5.7.1) directly and are the form in which radial matching conditions are usually written.

# Elementary functions at half-integer order

Half-integer order is the exceptional case in which the Bessel family reduces to elementary functions. At  $n = 0$ , equation (5.7.1) is  $u'' + u = 0$ , so  $u = \sin x$  or  $-\cos x$ :

$$j_0(x) = \frac{\sin x}{x}, \quad y_0(x) = -\frac{\cos x}{x}, \quad h_0^{(2)}(x) = j_0 - iy_0 = \frac{ie^{-ix}}{x}.$$

The recurrences of the parent functions supply the higher orders, e.g.

$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \quad y_1(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x}.$$

**Far field.** From the Hankel asymptotics of the parent functions,

$$h_n^{(1)}(x) \sim \frac{(-i)^{n+1}e^{ix}}{x}, \quad h_n^{(2)}(x) \sim \frac{i^{n+1}e^{-ix}}{x}, \quad x \rightarrow +\infty. \quad (5.7.4)$$

The  $1/x$  envelope is spherical spreading, the analogue of the  $1/\sqrt{x}$  cylindrical envelope of  $H_m^{(1,2)}$ .

**Convention flag.** With the time convention  $e^{-i\omega t}$  used in this chapter,  $h_n^{(1)}$  (and  $H_\nu^{(1)}$ ) is the outgoing wave. Much of the applied-scattering literature uses  $e^{+i\omega t}$ , where the outgoing role passes to  $h_n^{(2)}$  and  $H_\nu^{(2)}$ : always pair a Hankel label with the stated time convention.

# Elementary functions at half-integer order (cont.)

**Small argument.** For fixed  $n$  and  $x \rightarrow 0^+$ , the leading series terms of  $J_{n+1/2}$  and  $Y_{n+1/2}$  give

$$j_n(x) \sim \frac{x^n}{(2n+1)!!}, \quad y_n(x) \sim -\frac{(2n-1)!!}{x^{n+1}},$$

with double factorials  $(2n+1)!! = 1 \cdot 3 \cdot 5 \cdots (2n+1)$  and the standard empty-product convention  $(-1)!! = 1$ .

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# Formula sheet: $J_m$

For integer  $m \geq 0$ ,

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0,$$

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m},$$

$$J_{-m}(x) = (-1)^m J_m(x).$$

The generating function, valid for  $t \neq 0$ , is

$$e^{\frac{x}{2}(t-t^{-1})} = \sum_{m=-\infty}^{\infty} J_m(x) t^m.$$

# Formula sheet: $J_m$ (cont.)

The recurrences are

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x), \quad x \neq 0,$$

$$2J'_m(x) = J_{m-1}(x) - J_{m+1}(x),$$

$$J'_m(x) = J_{m-1}(x) - \frac{m}{x} J_m(x), \quad x \neq 0,$$

$$J'_m(x) = \frac{m}{x} J_m(x) - J_{m+1}(x), \quad x \neq 0.$$

The preferred integral representations are

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta, \quad m \geq 0,$$

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z},$$

with the optional contour form

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}.$$

## Formula sheet: $J_m$ (cont.)

For noninteger  $\nu$  the contour opens into the Hankel contour (5.3.4), equivalently into Sommerfeld's contour  $C$  (5.3.6):

$$J_\nu(x) = \frac{1}{2\pi} \int_C e^{i[x \cos z + \nu(z - \pi/2)]} dz, \quad x > 0.$$

For fixed  $m$ ,

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m \left[1 - \frac{x^2}{4(m+1)} + O(x^4)\right], \quad x \rightarrow 0,$$

$$J_m(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}), \quad x \rightarrow +\infty.$$

# Formula sheet: $Y_m$ , Hankel, Fourier–Bessel

For noninteger  $\nu$ ,

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)},$$

and for integer  $m$ ,

$$Y_m(x) = \lim_{\nu \rightarrow m} \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}, \quad x > 0.$$

The normalization is

$$W[J_m, Y_m] = \frac{2}{\pi x}.$$

For fixed integer  $m \geq 1$ ,

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad x \rightarrow 0^+,$$

whereas

$$Y_0(x) = \frac{2}{\pi} \left[ \log\left(\frac{x}{2}\right) + \gamma \right] + O(x^2 |\log x|).$$

For fixed  $m$  and  $x \rightarrow +\infty$ ,

$$Y_m(x) = \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}).$$

# Formula sheet: $Y_m$ , Hankel, Fourier–Bessel (cont.)

The Hankel functions are

$$\begin{aligned}H_m^{(1)} &= J_m + iY_m, & H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \\H_m^{(2)} &= J_m - iY_m, & H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}),\end{aligned}$$

with  $\chi_m(x) = x - m\pi/2 - \pi/4$ . The Hankel Wronskians follow from  $W[J_\nu, Y_\nu] = 2/(\pi x)$ :

$$W[J_\nu, H_\nu^{(1)}] = \frac{2i}{\pi x}, \quad W[J_\nu, H_\nu^{(2)}] = -\frac{2i}{\pi x}.$$

Half-integer orders give the spherical functions

$$z_n(x) = \sqrt{\frac{\pi}{2x}} Z_{n+1/2}(x), \quad z \in \{j, y, h^{(1)}, h^{(2)}\}, \quad Z \in \{J, Y, H^{(1)}, H^{(2)}\},$$

with  $h_n^{(1)}(x) \sim (-i)^{n+1} e^{ix}/x$  and  $h_n^{(2)}(x) \sim i^{n+1} e^{-ix}/x$ ; e.g.  $j_0 = \sin x/x$  and  $h_0^{(2)} = ie^{-ix}/x$ .

# Formula sheet: $Y_m$ , Hankel, Fourier–Bessel (cont.)

For the positive zeros  $\alpha_{m,j}$  of  $J_m$ ,

$$\int_0^1 J_m(\alpha_{m,j}x) J_m(\alpha_{m,k}x) x \, dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk},$$

with the Fourier–Bessel coefficients (5.6.10)–(5.6.11).